

CMSC3621 Data Structures and Algorithms Lab

Lab Assignment 4

Overview:

This assignment is for the students to review about inserting a data and removing a data in Binary Search Tree.

Details

In the attached .zip file, we provide a C++ project, where a `BinaryTreeNode` class and a `BinaryTree` class have been implemented. An additional `main.cpp` is used to test the two classes.

You need to complete a `BSTree` (Binary Search Tree) class in the `BSTree.h` file. In particular, this class inherits from the `BinaryTree` class. Thus, the parent functions such as

- `PreOrderTraverse(BinaryTreeNode<T> *root)`
- `InOrderTraverse(BinaryTreeNode<T> *root)`
- `PostOrderTraverse(BinaryTreeNode<T> *root)`
- `LevelOrderTraverse(BinaryTreeNode<T> *root)`

can be reused by a `BSTree` class instance.

The `BSTree` class in the `BSTree.h` file is incomplete. But we implement a `search()` function as an example. You need to complete other member functions, i.e. `insert()` and `remove()`. Please note the `search()` function has never been tested, you are responsible for fixing any possible problems.

You also need to modify the `main.cpp` file to test your `BSTree` class. Both `insert()` and `remove()` functions must be tested. More concretely, after applying insertions, you are to call `InOrderTraverse()` and `LevelOrderTraverse()` to output the tree. After applying removals, you should take the same actions.

Submission

- 1) BSTree.h file //Comments are needed to illustrate how do you implement the code
- 2) main.cpp file //Comments are needed to illustrate how do you implement the code
- 3) A report (in a PDF file), in the report you need to explain / provide
 - 3.1) how do you design and use the main.cpp file to verify that your BSTree class does work in a correct way
 - 3.2) screenshots and descriptions about how do you test your implementation

Submit all the three files above to D2L.

Important requirements:

- 1) Don't include any other extra files (that are not aforementioned).
- 2) Don't modify BinaryTreeNode.h and BinaryTree.h, otherwise your code may not be able to run in the grader's environment.
- 3) You only can write your code at certain places which are indicated by the existing comments
- 4) Any violation against the rules above will lead to extra penalty. If you missed the report, your submission will not be evaluated.

Rubric

- 1) 45 pts
- 2) 40 pts
- 3) 15 pts